



RAM Analysis

500

Dependability Analysis

ELEKTRA 2.x

LockTrac 6131

3BU 15000 4997 DUAPA / 06 RL

© GTS Ground Transportation Systems Austria GmbH

Entity Identifier	Document Identifier	Document Type Code	Revision
16282-9950	3BU 15000 4997 DUAPA	500	06 RL



Editions and Change History

Edition	Date	Written by	Approved by	Change Description
01	2008-08-13	TRSS-AT / RLR RAM Manager Markus Scheuer	TRSS-AT / RTSD TPM Erwin Glaser	See 1.3.1
02	2010-05-31	TRSS-AT / GRR RAM Manager Markus Scheuer	TRSS-AT / ERS TPM Erwin Glaser	See 1.3.2
03	2010-08-30	Thales Austria / GRR RAM Manager Markus Scheuer	Thales Austria / ERS TPM Erwin Glaser	See 1.3.3
04	2012-04-25	Thales Austria / GRR RAM Manager Markus Scheuer	Thales Austria / ERS TPM Erwin Glaser	See 1.3.4

Edition	Date	Written by	Verified by	Approved by	Change Description
05	2021-11-11	Thales Austria / TRR RAM-AM Markus Scheuer	Thales Austria / TQA QAM Denise Kovac	Thales Austria / TEPR SEM Erwin Glaser	See 1.3.5
06	2022-09-31	GTS Austria / TRR RAM-AM Florian Moser	GTS Austria / TQA QAM Denise Kovac	GTS Austria / TEPR SEM Erwin Glaser	See 1.3.6
		Digitally signed by MOSER Florian DN: o=Thales, cn=MOSER Florian, email=Florian.MOSER@thalesgroup.com, 0.9.2342.19200300.100.1.1=T0137474 Date: 2022.09.28 10:25:46 +02'00'	Digital unterschrieben von KOVAC Denise DN: o=Thales, cn=KOVAC Denise, email=denise.kovac@thalesgroup.com, 0.9.2342.19200300.100.1.1=T0143998 Datum: 2022.10.07 13:52:23 +02'00'	Digitally signed by GLASER Erwin DN: o=Thales, cn=GLASER Erwin, email=erwin.glaser@thalesgroup.com, 0.9.2342.19200300.100.1.1=T0071715 Date: 2022.10.18 08:24:41 +02'00'	



Table of Contents

1	SCOPE	5
1.1	Identification	5
1.2	Document Overview	5
1.3	History	6
1.3.1	Ed 01 RL	6
1.3.2	Ed 02 RL	6
1.3.3	Ed 03 RL	6
1.3.4	Ed 04 RL	6
1.3.5	Ed 05 RL	6
1.3.6	Ed 06 RL	7
2	REFERENCES AND TERMS	8
2.1	Referenced Documents	8
2.2	Standards and Procedures	9
2.3	Terms, Abbreviations and Acronyms	9
3	SYSTEM OVERVIEW	12
4	BASICS	14
4.1	Failure rates of components	14
4.2	Calculation of system dependability quantities	14
4.3	Fault tree analysis	14
4.4	Logical Equations	14
4.5	Failure rates and availability of redundant systems	15
5	FAILURE RATES	16
6	SUBSYSTEM ANALYSIS	18
6.1	Communication media	18
6.2	MMI	19
6.3	CC subsystem	19
6.4	FEC	20
6.4.1	General	20
6.4.2	EC/EC-MUX subsystem	20
6.4.3	IC subsystem	20
6.4.4	Remote EC/EC-MUX subsystem	20
6.5	LDC subsystem	21
6.5.1	RCU/SCU	21
6.5.2	TCD	22
6.5.3	IAU	22
6.6	SEA subsystem	23
6.7	No wear parts	23
6.8	Calculations	23
6.9	Results of the analysis	25
7	SYSTEM ANALYSIS	25
7.1	Results of the analysis	26
7.2	Fulfillment of requirements	27



Figures

Figure 1: System architecture of ELEKTRA 2.x	12
Figure 2: System architecture for high availability	13
Figure 3: Cabling of ELEKTRA 2.x with two-fold redundant LAN.....	18
Figure 4: System overview remote EC (ELEKTRA AT)	21

Tables

Table 1: Solution Identification	5
Table 2 : Referenced Documents.....	9
Table 3 : Standards and Procedures.....	9
Table 4 : Terms, Abbreviations and Acronyms.....	11
Table 5: Operators in logical equations	15
Table 6: MTBF and availability of redundant systems.....	15
Table 7: Failure rates.....	17
Table 8: Subsystems - Logical equations and failure rates.....	24
Table 9: Subsystems - MTBF and availability	25
Table 10: System - MTBF and availability	27
Table 11: LDC - MTBF and availability	27
Table 12: Availability classes for ELEKTRA 2.x	27
Table 13: Availability requirements for ELEKTRA 2.x	28



1 SCOPE

The objective of this document is to predict the failure behaviour of ELEKTRA 2.x in the field. Focus in this document is on ELEKTRA 2.x with the *conventional outdoor element controlling* based upon EC. Reliability aspects of the architectural alternative SEA ("Sichere Elektronische Außen-elementansteuerung") are considered in [RAM_ANA_SEA].

1.1 Identification

Project Name:	ELEKTRA System Version PV GEN 2.19.1
Project Id.:	100015491
Solution:	ATRENDDB13 GenSys 2.19.x
Product Identification:	ELEKTRA System Version PV GEN 2.19.0
Contract:	Internal Product Line Engineering Project
Customer: (internal/external)	EL AT, EL CH, EL INT
End User:	All countries
SIL:	SIL 4

Table 1: Solution Identification

1.2 Document Overview

This document is directed to the following project roles:

- Project / Product Design Authority (PDA)
- Product Line Manager (PLM)
- Solution Engineering Manager (SEM)
- System Architect (ARC)
- Requirements Manager (RM)
- RAM Assurance Manager (RAM-AM)
- Safety Assurance Manager (SAM)
- Quality Assurance Manager (QAM)
- IVVQ Manager –Validator (IVVQ-VAL)
- IVVQ Manager – Verification (IVVQ-VER)

If necessary, this document is updated and made available in a new edition. It can be found in ClearCase under version management (VOB 'DOCUMENTATION') and it can be called via DWF (section 'ELEKTRA').



1.3 History

1.3.1 Ed 01 RL

- This edition is based upon and successor of following documents
Dependability Analysis, ANV: 3BU 15000 0671 PEAPA, Ed. 01 and
Dependability Analysis; ANV: 3BU 00134 0000 BCAPA, Ed. 01.

Change information below refers to these two documents.

- Reliability calculations for D6 added.
- Failure Rate for IMO added.
- Minor corrections in LDC calculation in *Table 9: Subsystems - MTBF and availability*.
- MTTR (CC switchover time) changed from 2 minutes to 3 minutes according to DPR:0203 from [EL2_SYS_SRS].
- Definition section 2.3 updated
- MTBF and availability data somewhat more detailed and collected in *Table 9: Subsystems - MTBF and availability* and *Table 10: System - MTBF and availability*.
- Minor updates according to review comments

1.3.2 Ed 02 RL

- Long distance communication (LDC) via RCU/SCU added in chapter 6.5
- LDC examined by its own in chapter 7
- Tables in chapter 6.6, 6.9 and 7 updated
- References updated

1.3.3 Ed 03 RL

- Availability aspects: *Communication ELEKTRA to AzLM via OCS* added in chapter 7.2
- Subsection 4.5: *Failure rates and availability of redundant systems* added
- Section 7.1: *Results of the analysis* updated
- Additional remarks about availability in section 7.2: *Fulfillment of requirements*.
- Reference section updated
- Review comments incorporated

1.3.4 Ed 04 RL

- Upgrade due to handling of remote EC
- EC-MUX introduced with references to [RAM_ANA_SEA]
- References updated
- Review comments incorporated

1.3.5 Ed 05 RL

- MEN-D10 added
- MMI subsystem variants extended
- LDC subsystem variants extended
- IxL – AzLM connection now considered via OCS-SB



- Document references updated
- Small section 6.7 addressing wear parts added
- References to dependability requirements added
- Some minor corrections/improvements
- New document template

1.3.6 Ed 06 RL

- New document template
- Document references updated
- LDC subsystem variants extended
 - X25 over UDP/IP with RaSTA communication added in section 6.5.1
 - X25 over UDP/IP with RaSTA communication added in Table 8, Table 9 and Table 11
- Some minor corrections/improvements



2 References and Terms

2.1 Referenced Documents

Id	Reference
[ARPR_PSD_D10]	AlcaRel Reliability Prediction Report; 3BU 15651 AAAA HAAPC, Ed.01 Netzgerät D10 Powertronic; Power Supply; 3BU 15651 AAAA DUAPA, Ed. 01
[EL2_remEC_ADD]	Architectural Design Document – Ausgelagerte ECs, mehrere Stromversorgungen; 3BU 15000 2191 DSAPC, Ed. 04
[EL2_CH_IAU_IXL_RBC]	Architectural Design Document – Technical Functionality - IF IXL - RBC; 3BU 15000 1704 PEAPA, Ed. 04
[EL2_CH_IF_IXL_RBC]	Architectural Design Document - IF IXL-RBC over OCS; 3BU 15000 3019 DSAPC, Ed. 15
[EL2_CH_REQ_TCD]	ELEKTRA – SBB CFF FFS – Systemanforderungen TCD; 3BU 00426 0000 DTAPC, Ed. 02
[EL2_SEA_ADD]	Architectural Design Document - SEA, ELEKTRA 2.x ÖBB; 3BU 61400 0004 DSAPA, Ed. 06
[EL2_SYS_ADD]	Architectural Design Document – System Architecture; 3BU 15000 0400 PEAPA, Ed. 24
[EL2_SYS_LIM_AT]	Systemgrenzen ESTW; ELEKTRA ÖBB; 3BU 15000 0203 EDAPC, Ed. 11
[EL2_SYS_LIM_CH]	System Subsystem Design Description; Statische und dynamische Systemgrenzen; ELEKTRA Schweiz; 3BU 15000 3435 DSAPC, Ed. 03
[EL2_SYS_LIM_INT]	System Description; System Limits ELEKTRA International 3BU 15000 9021 EDAPA, Ed. 12
[EL2_SYS_SRS]	System Requirements Specification; 3BU 15000 0300 DTAPA, Ed. 39
[IF_EL2_AzLM_ADD]	Architectural Design Document – IF Elektra-AzLM over OCS-SB; 3BU 15000 5138 DSAPC, Ed. 06
[RAM_ANA_D10]	Dependability Analysis; Core System D10; 3BY 09015 0087 DUAPA, Ed. 02
[RAM_ANA_D6]	RAM Analysis; Core System D6; 3JA 80001 AAAA VFAPA, Ed. 03
[RAM_ANA_ELINT]	Dependability Analysis; ELEKTRA 2 – International 3BU 15000 9101 DUAPA, Ed. 03
[RAM_ANA_SEA]	ELEKTRA 2.x; Dependability Analysis – SEA; 3BU 61400 0037 DUAPA, Ed 03
[RAM_PLAN_GEN]	RAM Plan Generic; 3BU 15000 1712 DUAPA, Ed. 04
[THALES_FAILRATE]	Reliability Standard Failure Rate Factor; 1AA 00210 0001 BCZZA, Ed. 04



Table 2 : Referenced Documents

2.2 Standards and Procedures

Id	Reference
[Birolini]	Reliability Engineering. Theory and Practice; Ed. 08 Alessandro Birolini; ISBN: 3662571897
[DOC_RULES]	Thales Austria Document Management Rules, Chorus 2 83530061-DOC-AUT-EN, Rev. 010
[EN 50126_1:2017]	EN 50126-1:2017: Railway Applications. The Specification and Demonstration of Reliability, Availability, Maintainability and Safety (RAMS). Generic RAMS Process
[IEC60050(192)]	IEC 60050-192:2015: Ed. 1.0; International Electrotechnical Vocabulary. Part 192: Dependability
[MIL_HDBK217F]	US MIL-HDBK 217F, Reliability prediction for electronic equipment; Version F, 1991; 1AA 00230 0001 ASZZA, Ed.01

Table 3 : Standards and Procedures

2.3 Terms, Abbreviations and Acronyms

Term	Meaning
ARAMIS	A dvanced R ailway A utomation and M anagement I nformation S ystem
Availability	Probability that an item is in a state to perform a required function under given conditions at a given instant of time, assuming that the required external resources are provided. [IEC60050(192)] A qualitative definition, focused on <i>ability</i> rather than on <i>probability</i> is also usual. The <i>steady-state value</i> of the point availability can be expressed in general by $\frac{MTTF}{MTTF + MTTR}$. [Birolini]
BP	B ack P anel
CB	C arrier B oard
CC	C entral C ontroller
CF	C ompact F lash C ard
CM	C ommunication M edia
CPU	C entral P rocessing U nit
DGP	D iagnostic P rocessor
DPR	D ependability R equirement
EC	E lement C ontroller
EC-MUX	E lement C ontroller M ultiplexer
ESM	E lectronic S torage M edium



Term	Meaning
Failure rate	Limit for $\delta \rightarrow 0$ of the probability that an item will fail in the time interval $(t, t + \delta]$, given that the item was new at $t = 0$ and did not fail in the interval $(0, t]$, divided by δ . [Biolini]
Fault tree	A logic diagram showing which fault modes of subitems or external events, or combinations thereof, result in a given fault mode of the item. [IEC60050(192)]
FEC	Field Element Controller
FiT	Failures in Time (unit: $10^{-9}h^{-1}$)
Ground Benign (GB)	Nonmobile, temperature and humidity controlled environments readily accessible to maintenance; includes laboratory instruments and test equipment, medical electronic equipment, business and scientific computer complexes, and missiles and support equipment in ground silos. [MIL_HDBK217F]
GB 40	Ground Benign 40° C
GRAULI	Graphic Automatic Light
HDLC	High-Level Data Link Control
IAU	Interface Adaption Unit
IC	Interface Controller
IC1D	Interface Controller 1D
ILTIS	Integriertes LeitTechnik- und Informations-System (Siemens)
IMO3D	Input Module 3D
LAN	Local Area Network
LDC	Long Distance Communication
[MIL_HDBK217F]	The Military Handbook is published by the US Department of Defense, and provides mathematical models for electrical, electronic and electro-mechanical components. Each model calculates the component failure rate for given combinations of environmental conditions, component value and type, electrical stress conditions and manufactured quality level
MDS	Monitoring and Data Collection System
MMI	Man Machine Interface
MTBF	Mean Time Between Failures - The expectation of the operating time between failures. [IEC60050(192)] $MTBF$ is usually reserved for items with <i>constant</i> failure rate λ . In this case, $MTBF = 1/\lambda$ is the expected value (mean) of the <i>exponentially distributed item's failure-free operating time</i> . In particular $MTBF \approx T/k$, where T is the given, <i>fixed</i> cumulative operating time and k the total number of failures (failed items) during T . [Biolini]
MTTR	Mean Time To Restore - The expectation of the time to restoration. [IEC60050(192)] $MTTR$ is the expected value (mean) of an item's repair time. It is obtained from the distribution function $G(t)$ of the repair time as $MTTR = \int_0^{\infty} (1 - G(t))dt$. For reliability or availability calculation a constant repair rate μ (i.e.



Term	Meaning
	exponentially distributed repair times with $\mu = 1/MTTR$) can often be used to get approximate results. [Biolini]
NAC	Network Adapter Card
OCS	One Channel Safe
OEC	Outdoor Element Controller
PAD	Packet Assembler Disassembler
PBA	Printed Board Assembly
PMO	Point Module
PSD	Power Supply Device
RAM	Reliability, Availability and Maintainability
RaSTA	Rail Safe Transport Application
RBC	Radio Block Center
RIC	Remote Interface Controller
RCU	Remote Controller Unit
RZÜ	Rechnergesteuerte Zugüberwachung
SCU	Serial Communication Unit
SEA	Sichere Elektronische Außenelementansteuerung
SSU	Schnittstellenumsetzer
TA CT04	Transition Adapter CT04
TCD	Transmission Controller Double Channel
TDT	Total Down Time
TTP	Time Triggered Protocol
uSD	Micro Secure Digital memory card
WebRide	Web-based Remote Interlocking Diagnostic Engine
WP	WorkPlace

Table 4 : Terms, Abbreviations and Acronyms



3 System Overview

A description of the ELEKTRA 2.x system can be found in [EL2_SYS_ADD]. The system consists of the following units:

- CC subsystem
- FEC with up to eight EC
- MMI with up to ten workplaces
- ESM, MDS¹, radio clock
- LDC subsystem
- DGP
- IC Interface to field elements.
- SEA (alternative to conventional outdoor element controlling)
- EC/EC-MUX subsystems remotely installed

The graphic from [EL2_SYS_ADD] below shows the system architecture of ELEKTRA 2.x.

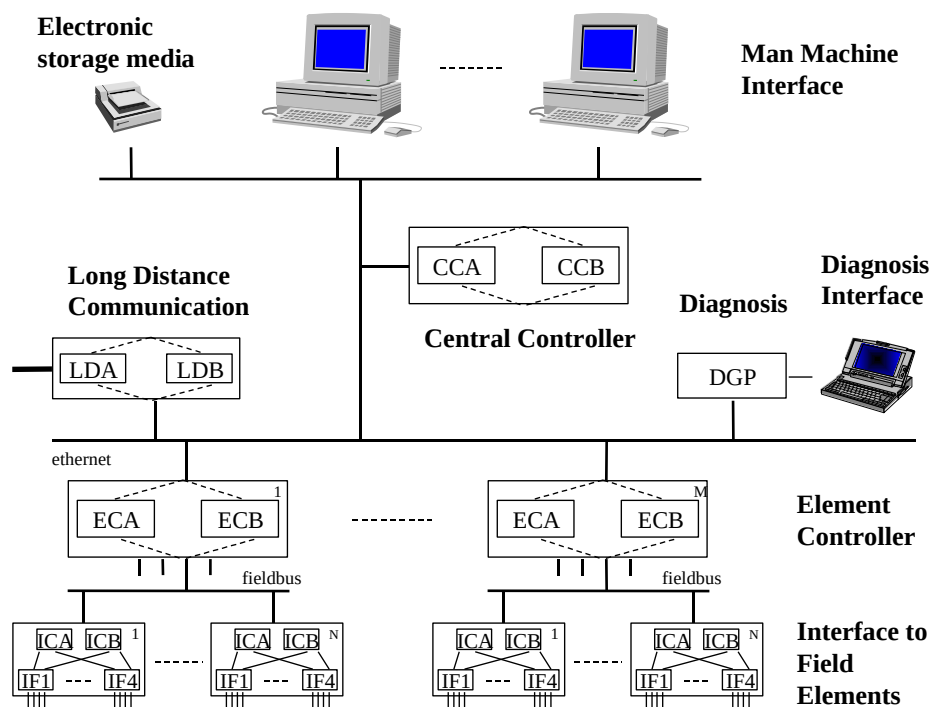


Figure 1: System architecture of ELEKTRA 2.x

¹ Successor of WebRide



An example configuration of ELEKTRA 2.x for high availability needs from [EL2_SYS_ADD] is given below.

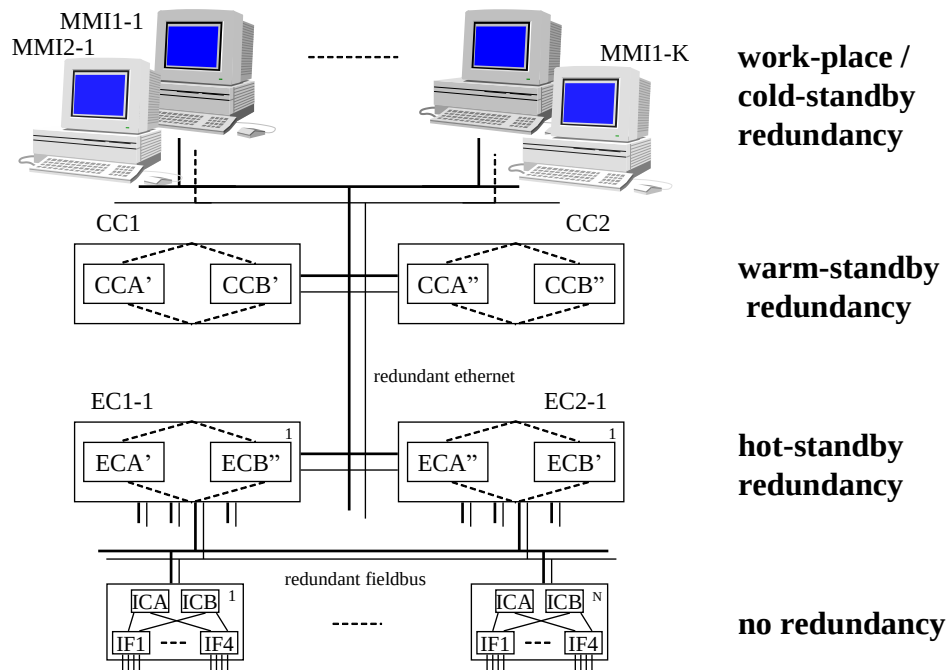


Figure 2: System architecture for high availability

Due to simplicity reasons only one EC component (including the corresponding IC components) is shown in figure 2. MMI have two redundancies. But it is also possible to use only one redundancy for all MMI (see 6.2).



4 BASICS

4.1 Failure rates of components

The failure rates of components were calculated by following a THALES standard method described in [THALES_FAILRATE]. This method is principally based on the models of [MIL_HDBK217F]. In order to avoid too conservative data, a corrective factor is applied reflecting the real failure behaviour in field.

For the ELEKTRA 2.x system a *ground, benign* environment is assumed. Sometimes, in case of similar HW-variants, the representative with the most conservative failure rate was used.

Only randomly occurring hardware failures are taken into account. Faults of a systematic nature are minimised by following the rules of CENELEC during the development process according to [RAM_PLAN_GEN].

4.2 Calculation of system dependability quantities

The calculation of the overall failure rate of PBA and devices is done by summing up all component failure rates located on it (parts stress method according to [MIL_HDBK217F]).

Calculations of system quantities are based on the constant failure rates of PBA and devices and an average repair time (MTTR) of 8 hours.

4.3 Fault tree analysis

The calculations were performed using fault tree analysis techniques and are based on so-called logical equations. The logical equations represent all possible failure combinations, which can lead to a system failure. Starting with the undesired top event all successive subordinate events are identified until the basic events are reached. The basic events are failure events that cannot further be divided, without limiting further development. This means that those events are the failures of individual PBA and devices, respectively. At this level the failure behaviour of an item is derived from the constant failure rate assigned.

4.4 Logical Equations

The logical equations use operators to combine failure events from simple components to failure events of more complex components. The operators are described in the table below. The system failure rate λ_{sys} denotes the failure rate calculated from the sub-system failure rates $\lambda_{1,2}$ and their corresponding repair rates $\mu_{1,2}$.



Meaning	Example	Failure rate λ_{Sys}
+ denotes logic OR	$comp_{Sys} = comp_1 + comp_2$	$\lambda_{Sys} = \lambda_1 + \lambda_2$
* denotes logic AND	$comp_{Sys} = comp_1 * comp_2$	$\lambda_{Sys} = \frac{\lambda_1 \lambda_2 (\mu_1 + \mu_2)}{\mu_1 \mu_2}$
$k comp_1 + \dots + comp_n$ denotes " k out of n "	$comp_{Sys} =$ $< k comp_1 + \dots + comp_n >$	$\lambda_{Sys} \approx k \lambda \binom{n}{k} \left(\frac{\lambda}{\mu} \right)^{n-k}$
1 out of 2	$< 1 comp_1 + comp_2 >$	$\lambda_{Sys} \approx 2 \frac{\lambda^2}{\mu}$
2 out of 3	$< 2 comp_1 + comp_2 + comp_3 >$	$\lambda_{Sys} \approx 6 \frac{\lambda^2}{\mu}$

Table 5: Operators in logical equations²

Postfixes "-x", $x = 1, \dots, n$ are used to indicate multiplicity of units. For further detailed denotation the underline "_" is used. In the LDC subsystem "-l" stands for the local station part and "-r" stands for the remote station of a single line.

We do not consider wearing parts in the logical equations. We assume that all parts have a constant failure rate during their useful life phase.

4.5 Failure rates and availability of redundant systems

The table below lists MTBF and availability formulas for non-redundant and redundant systems.

System	$MTBF_{Sys}$	PA_{Sys}
1oo1 non-redundant	$\frac{1}{\lambda}$	$1 - \frac{\lambda}{\mu}$
1oo2 hot standby	$\frac{\mu}{2\lambda^2}$	$1 - \frac{2\lambda^2}{\mu^2}$
1oo2 warm standby	$\frac{1}{\lambda}$	$1 - \frac{\lambda}{\mu_{so}}$
2oo3 hot standby	$\frac{\mu}{6\lambda^2}$	$1 - \frac{6\lambda^2}{\mu^2}$

Table 6: MTBF and availability of redundant systems

² Formulas listed here can be found e.g. in [Birolini].



Legend:

$MTBF_{Sys}$	MeanTime Between Failures of the System
PA_{Sys}	Point Availability of the system
λ	Failure rate of a subsystem
μ	Repair rate (manual exchange) of a subsystem
μ_{SO}	Repair rate (automatically SwitchOver) of a subsystem

5 Failure rates

Failure rates are either calculated by the Thales method or figures were taken from manufacturer's data sheets. In some cases, estimated failure rates are stated. They typically represent a conservative approach based on calculations of similar items using the Thales method. In order to consider the experience with the Thales method compared with conservative approaches from [MIL_HDBK217F], some devices have been adapted.

Item	Part Number	FIT [1e-09/h] GB 40	Origin
MEN-D2	3CR 30105 AAAA 3CR 30105 ABAA	1 560	Thales method
MEN-D6	3BU 15585 AAAA 3BU 15585 ABAA	3 780	Thales method ³
MEN-D10	3BU 15650 AAAB	3 115	Thales method ⁴
CF	3JA 80250 AAAC 3JA 80250 BAAA	787	Thales method
uSD	7HA 10080 AAAA	333	Manufacturer
TA CT04	3CR 30309 AAAA	231	Thales method
PSD MEN-D2, MEN-D6	3CR 30604 AxAA	1 200	Estimation
PSD MEN-D10	3BU 15651 AAAA	626	Thales method ⁵
Backpanel	3CR 06012 AAAA 3CR 06013 AAAA	1 580	Thales method
Switch	3BU 15112 ABAA 3BU 15112 ABAB	2 200	Estimation
TTP-M Module	3CR 30307 AAAx, 3CR 31020 AAAx	270	Thales method
M-Module Carrier Board	3BU 15594 AxAA	750	Thales method
HDLC M-Module	3CR 30312 AAAx	420	Thales method
KVM CPU Switch Multiscreen	3BU 15114 AAAA	2 000	Estimation
KM CPU Switch	3BU 15115 AAAA 3BU 15115 ABAA	2 000	Estimation
PAD Multiprotocol Switch	3BU 15326 AAAA 3BU 15327 AAAA 3BU 15328 AAAA	2 000	Estimation

³ [RAM_ANA_D6]

⁴ [RAM_ANA_D10]

⁵ [ARPR_PSD_D10]



Item	Part Number	FIT [1e-09/h] GB 40	Origin
Modem	3BU 15110 0344	3 000 ⁶	Estimation
	3BU 15110 0345		
	3BU 15110 0512		
	3BU 15110 0513		
	3BU 15110 0275		
	3BU 15110 0537		
	3BU 15110 0538		
	3BU 15110 0638		
	3BU 15110 0639		
	3BU 15110 0640		
	3BU 15110 0641		
IC1D	3BU 15507 AAAA 3BU 15541 AAAA	2 720	Thales method
IMO3D	3BU 15612 AAAA 3BU 15612 AAAB	2 642	Thales method
PMO	3BU 15490 ACAA 3BU 15490 ADAA 3BU 15490 AEAA 3BU 15490 AF AA	1 856	Thales method
MMI	3BU 15111 DAAA 3BU 15111 DEAA	10 000	Estimation

Table 7: Failure rates

⁶ The manufacturer's stated high failure rates (15 000 FiT – 16 500 FiT) seem to come from plain [MIL_HDBK217F] and are not confirmed in field. We stick to the estimation of 3 000 FiT.



6 Subsystem analysis

In the sections below subsystems with different kind of redundancy characteristics are employed. They are

- non-redundant (one MEN-D2/MEN-D6/MEN-D10 board, one PSD)
- warm standby redundancy (two MEN-D2/MEN-D6/MEN-D10 boards in parallel, two PSD)

With respect to *reliability* (MTBF) the warm standby configuration has essentially the same behavior as a non-redundant system. This follows, because a failure of the active component causes a non-negligible interrupt of the subsystem.

But, with respect to *availability* the subsystem is a twofold redundant system, with a short switchover time (a few minutes).

- hot standby redundancy (two MEN-D2/MEN-D6/MEN-D10 boards in parallel, two PSD)

6.1 Communication media

The ethernet LAN is used set up by switches connecting all devices together to form a collision domain. Since there is a redundant LAN, two switches creating two independent collision domains are required. The switches are hot-pluggable, so that nodes can be replaced individually without disrupting other nodes (DPR:0207). The necessary cables are assumed to be perfect. The basic system (distance between the station room and the station manager room < 100 meters) only needs two switches. From the reliability point of view the communication media can be seen as a system consisting of two switches in parallel.

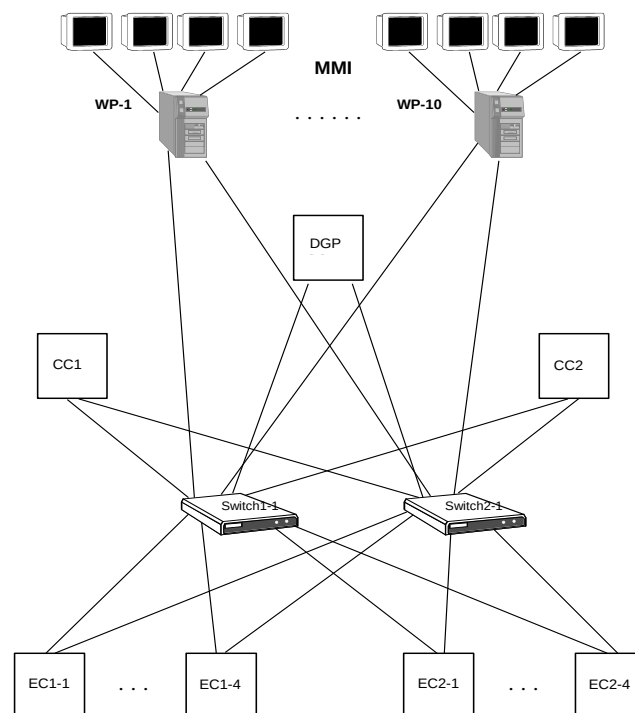


Figure 3: Cabling of ELEKTRA 2.x with two-fold redundant LAN

LDC links, ESM, MDS (WebRide) and radio clock are not shown in this figure.



6.2 MMI

Workplace redundancy may be used on the MMI layer, i.e. more than one MMI workplace can be used in parallel. In case of a faulty MMI component the user can move to another MMI workplace.

Alternatively, one MMI workplace can have two redundancies (two PCs). When using EBO2 (ELEKTRA AT and ELEKTRA International) both redundancies are running (hot-standby). In case of a HW-fault of the active PC a KVM CPU switchover unit can be handled manually to make the other running PC the active one avoiding usually operational impact.

In Switzerland a separated so-called emergency-workplace is provided in the ELEKTRA computer room of an interlocking station. This non-redundantly equipped workplace can be used as fallback solution in case the other ILTIS MMI workplaces do not work properly.

ELEKTRA International increases the number of available MMIs with a remote-control concept. Reliability aspects of this variant are considered in detail in [RAM_ANA_ELINT].

A workplace may fail because of the following: A single failure in

- NAC
- Hard disk
- PC hardware (like chipset, RAM, expansion slots, controllers, etc.)
- Graphic card
- USB-dongle (workplace identification)

and in case of a redundant MMI workplace:

- Hardware for switching over mouse, keyboard, monitor, loud speaker and power supply

We consider following characteristics:

- A single workplace without redundancy
Due to the fact that the hardware listed above is changed on a regular basis it is reasonable to make a conservative approach concerning the predicted failure behavior in the field. As a consequence, the MTBF of a PC - all components in series – is assumed to be $10^5 h$.
- Two workplace instances to improve reliability
If two workplaces are available we assume in case of an MMI failure at the active workplace a change to the other workplace without operational impact. This results in an MTBF of $6.25 * 10^8 h$ for a workplace and a TDT per year less than a minute.
- A redundant workplace with a manual switchover HW unit.
A redundant workplace is not available if both MMI PCs fail within the MTTR of 8 hours or the active MMI PC fails together with the switchover unit. Here we may assume that in case of an MMI PC fault no operational impact is given since the switchover activity can be performed in short time. The MTBF main contribution is given by the switchover unit resulting in an MTBF of $5.00 * 10^5 h$ and again a TDT per year less than a minute.

6.3 CC subsystem

In the CC subsystem a warm standby concept is given. The logical failure mechanism of this kind of redundancy can approximately be seen as a single element system with a failure rate of one MEN unit (one MEN-D2/MEN-D6/MEN-D10 board and the associated power supply device) and a down time of up to three minutes in mean (DPR:0203).

In order to assure high operational availability of the CC subsystem a failure of a switch or any fault of an active CC which only causes one Ethernet link to fail, does not lead to an automatic switchover to the standby CC (DPR:0212).



6.4 FEC

6.4.1 General

The FEC consists of the following subsystems

- EC/EC-MUX subsystem
- FEC bus
- IC subsystem
- HW interface boards

6.4.2 EC/EC-MUX subsystem

The EC/EC-MUX subsystem is equipped with “hot-standby” redundancy. This means that a single failure is tolerated and does not lead to any interruption of system parts (DPR:0218, DPR:0219). Further analysis is based on a variety of subsystem configurations from medium to maximum size. They range from four EC boards up to eight EC/EC-MUX boards.

The EC-MUX subsystem is part of the ELEKTRA 2.x SEA Subsystem and considered in [RAM_ANA_SEA]. From the reliability point of view the EC-MUX subsystem (indoor part of SEA) and the EC subsystem do not differ in their failure behaviour.

6.4.3 IC subsystem

The individual IC1D boards are not duplicated because a failure of one IC1D board affects only at most 3 peripheral elements (e.g. 3 points). A failure of a single IC does not affect remaining fault-free ICs (DPR:0217). Due to a similar severity level in case of faults the other IC subsystems PMO and IMO are also not duplicated.

6.4.4 Remote EC/EC-MUX subsystem

The remote EC (resp. EC-MUX) is located in a remote station. It provides control of the elements of the remote station while the vital ELEKTRA CC core in the central station is used for controlling the EC/EC-MUX subsystems of both stations, the local as well as the remote one.

Here we put the focus on EC subsystems. Handling of EC-MUX is analysed in [RAM_ANA_SEA]. From the reliability aspect the EC-MUX subsystem (indoor part of SEA) can be treated in the same way as the EC subsystem. The communication to remote EC is realized with two modems per LAN. The reliability figures for these *communication media remote* is stated and respected in section 6.8 and the following sections.

The figure below from [EL2_remEC_ADD] gives an example of a central CC with remote EC in two remote stations. Three columns show two remote stations, one on the left, one right side and the central station in the middle. The remote station on the left side comprises a remote EC (“EC x-4”), while the remote station on the right side has a remote EC (“EC x-5”) and a remote EC-MUX (“EC x-6”) installed⁷.

⁷ The two switches in the graphic left and right of the CC are in fact the two redundant switches of the ELEKTRA internal LAN.

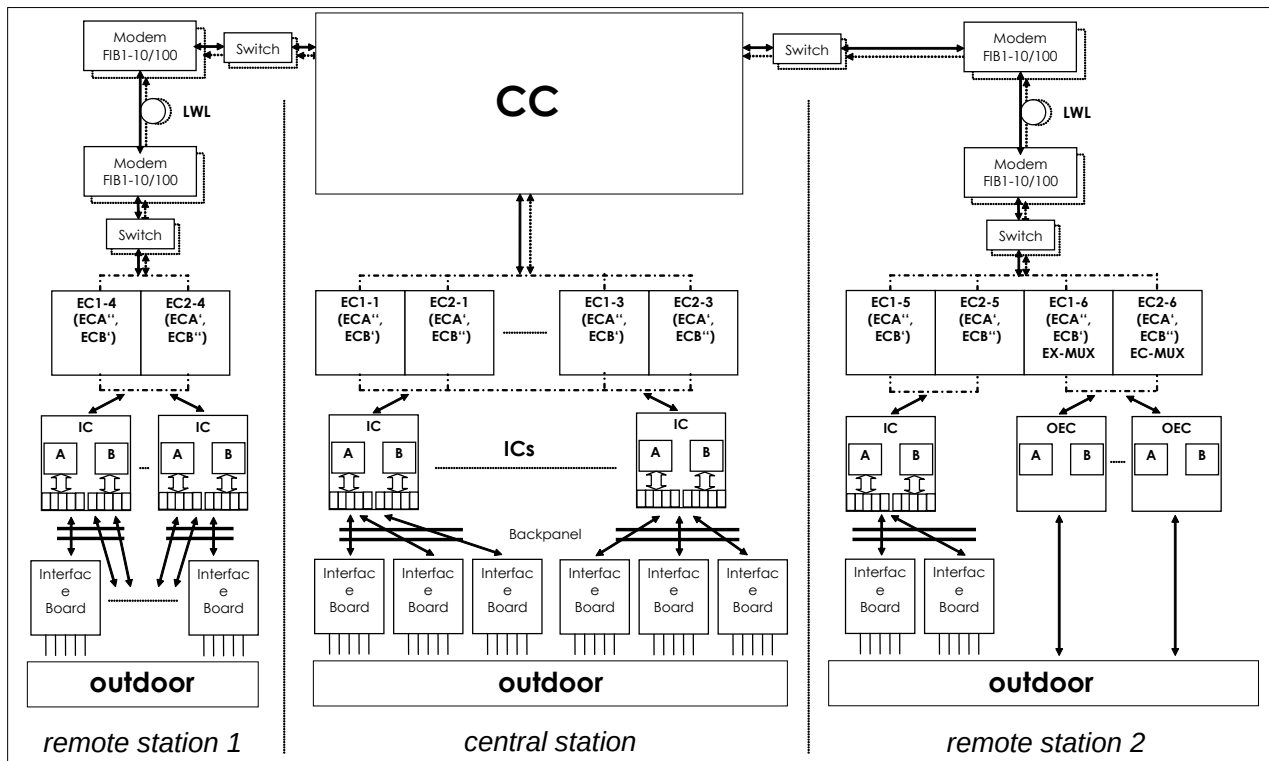


Figure 4: System overview remote EC (ELEKTRA AT)

Constraints:

- Currently the EC-MUX is capable of controlling light points but no other elements (points, axle counter, etc.). The other elements have to be controlled by remote EC (see [EL2_remEC_ADD] for more information).
- According to [EL2_SYS_LIM_AT] a maximum number of eight EC resp. EC-MUX can be controlled in ELEKTRA AT. In [EL2_SYS_LIM_CH] a maximum number of six EC resp. EC-MUX is stated for ELEKTRA CH. In [EL2_SYS_LIM_INT] theoretical limits for the maximum number EC and EC-MUX are stated, which have to be considered depending on some other limiting values used.
 - In the calculations below (see **Table 10**) configurations with up to eight EC resp. EC-MUX are considered.
- At least one EC has to be installed locally.

6.5 LDC subsystem

The focus of this RAM Analysis is the local station the ELEKTRA 2.x system is residing. Therefore, we consider the hardware components of only one side of an LDC subsystem.

The LDC realisations by RCU/SCU including RaSTA, TCD and IAU are presented in the sections below (see [EL2_SYS_ADD]).

6.5.1 RCU/SCU

The RCU is used to manage interfaces listed below. In order to fulfil high availability requirements, an RCU is designed as warm-standby subsystem executing on MEN-Dx HW. One or more SCU are used for communication between RCU and the remote system.



In case a primary RCU has a fault, a short operational interrupt (due to “Generalabfrage”) occurs until the secondary unit becomes active. If the secondary RCU fails it has no operational impact. If an active link of a redundant communication on an SCU fails this also might have a short operational impact.

In ELEKTRA AT an RCU can be connected with up to five SCU. An SCU can support up to four redundant or up to eight non-redundant links ([EL2_SYS_LIM_AT]). Corresponding limits for ELEKTRA CH and ELEKTRA International are stated in [EL2_SYS_LIM_CH] and [EL2_SYS_LIM_INT].

The general case is that only one SCU HW board is sufficient to support a limited number of connections. Here both RCU- and SCU-application can reside on one HW board, then called SSU. For redundancy reasons two boards are used, forming a warm-standby subsystem. Note, this also reduces the number of PBA which may fail.

A connection via X25 over UDP/IP with RaSTA can be used to replace the serial X.25 communication. The RaSTA interfaces are connected directly via the RCU. The most common case is that both RCU- and SCU-application reside on one HW board.

In ELEKTRA Switzerland RCU and SCU of the same channel are coupled due to safety reasons (see [EL2_CH_IF_IXL_RBC] for a detailed description). This means that a fault of one RCU or SCU implies a reset of the whole channel (RCU and SCU).

The interfaces supported by RCU/SCU are ([EL2_SYS_ADD]):

RCU/SCU used by ELEKTRA Switzerland for

- Interface to ILTIS
- Interface to RBC Thales

RCU/SCU used by ELEKTRA AT and ELEKTRA International for

- Remote control of other interlocking systems
- Block communication between electronic interlocking systems
- Interface to RIC
- Interface to Grauli
- Interface to RZÜ (Aramis)

RCU/SCU used by ELEKTRA AT for

- X25 over UDP/IP with RaSTA communication

RCU/SCU used by ELEKTRA International additionally for

- Interface to AKF (for customer Gysev)
- Interface to Köfe (used instead of RZÜ)

6.5.2 TCD

TCD is used by ELEKTRA Switzerland for

- Electronic block between 2 ELEKTRAs

Depending on the availability requirements of the connected subsystem the TCD is either non-redundant or hot-redundant in use ([EL2_CH_REQ_TCD]).

6.5.3 IAU

IAU is used by ELEKTRA Switzerland for



- Interface to RBC Alstom

The IAU is similarly to the CC subsystem designed as warm standby subsystem. Both IAU MEN-Dx units are connected to the CC and permanently updated with relevant data. But only the active unit has an open TCI/IP connection to the RBC. In case of a fault of the active unit, the CC demands a switchover and the other IAU unit becomes active establishing the connection to the RBC (see [EL2_CH_IAU_IXL_RBC]).

6.6 SEA subsystem

The dependability issues of the SEA subsystem of ELEKTRA are analyzed in [RAM_ANA_SEA]. They are not considered within this document.

6.7 No wear parts

In order to reduce life-cycle costs and conveniently support maintenance one of the design goals was to avoid preventive maintenance. For this reason no wear parts on PBA and devices are admissible (DPR:0225).

Also fans necessary to cool PBA in order to support longer life-times are installed in a *redundant* way. A fault of a single fan wheel does not influence the operation. Fans are monitored and a failure is indicated to the operator (DPR: 0216).

6.8 Calculations

The table below lists the logical equations for the sub-systems. The rightmost column of the table shows the corresponding failure rates.

Subsystem	Logical equations	FiT [1E-9/h] GB 40
CC	$CC_{D2_warm} = Men-D2 + CF + TA + PSD_CC + 1/4*BP$	4 173
	$CC_{D6_warm} = Men-D6 + CF + TA + PSD_CC + 1/4*BP$	6 393
	$CC_{D10_warm} = Men-D10 + uSD + TA + PSD_CC + 1/4*BP$	4 700
EC	$EC-n_{D2} = EC1-n_{D2} * EC2-n_{D2} \quad (n = 1..4)$	0.58
	$EC1-n_{D2} = MEN-D2_EC1-n + CF + TA + M-Module_CB_EC1-n + TTP-Mod(4x)_EC1-n + PSD_EC1-n + 1/4 * BP \quad (n = 1..4)$	6 003
	$EC-n_{D6} = EC1-n_{D6} * EC2-n_{D6} \quad (n = 1..4)$	1.08
	$EC1-n_{D6} = MEN-D6_EC1-n + CF + TA + M-Module_CB_EC1-n + TTP-Mod(4x)_EC1-n + PSD_EC1-n + 1/4 * BP \quad (n = 1..4)$	8 223
	$EC-n_{D10} = EC1-n_{D10} * EC2-n_{D10} \quad (n = 1..4)$	0.68
	$EC1-n_{D10} = MEN-D10_EC1-n + uSD + TA + M-Module_CB_EC1-n + TTP-Mod(4x)_EC1-n + PSD_EC1-n + 1/4 * BP \quad (n = 1..4)$	6 530
EC-MUX	$EC-MUX-n_{D6} = EC-MUX1-n_{D6} * EC-MUX2-n_{D6} \quad (n = 1..4)$	0.65



Subsystem	Logical equations	FiT [1E-9/h] GB 40
	$EC-MUX_{1-nD6} = MEN-D6_EC_{1-n} + CF + TA + PSD_EC_{1-n} + 1/4 * BP \quad (n = 1..4)$	6 393
	$EC-MUX_{1-nD10} = EC-MUX_{1-nD10} * EC-MUX_{2-nD10} \quad (n = 1..4)$	0.35
	$EC-MUX_{1-nD10} = MEN-D10_EC_{1-n} + uSD + TA + PSD_EC_{1-n} + 1/4 * BP \quad (n = 1..4)$	4 700
IC	IC = IC1D	2 720
	IMO = IMO3D	2 642
	PMO = PMO	1 856
MMI	WP ₁₀₀₁ = WP	10 000
	WP ₁₀₀₂ = WP ₁ * WP ₂	1.60
	WP _{1002_s0} = WP ₁ * WP ₂ + so_unit	2 002
	WP = NAC + Hard-disk + Graphic-Card + additional PC-HW	10 000
Comm. Media	CM = Switch_1 * Switch_2	0.08
Comm. Media remote	CM_rem ₁₀₀₂ = Switch ₁₀₀₂ + Modem ₁₀₀₂ + Modem_rem ₁₀₀₂ + Switch_rem ₁₀₀₂	0.44
LDC _{TCD}	$LDC_{D2_non-red} = MEN-D2 + PSD + M-Module-CB + HDLC-M-Module + Modem$	6 930
	$LDC_{D2_hot} = (LDC_{D2_non-red})_1 * (LDC_{D2_non-red})_2$	0.77
	$LDC_{D6_non-red} = MEN-D6 + PSD + M-Module-CB + HDLC-M-Module + Modem$	9 150
	$LDC_{D6_hot} = (LDC_{D6_non-red})_1 * (LDC_{D6_non-red})_2$	1.34
	$LDC_{D10_non-red} = MEN-D10 + PSD_{D10} + M-Module-CB + HDLC-M-Module + Modem$	7 911
	$LDC_{D10_hot} = (LDC_{D10_non-red})_1 * (LDC_{D10_non-red})_2$	1.00
LDC _{RCU}	$LDC_{D2_warm} = (SSU_{D2} + HDLC-CB + HDLC-M-Modul + PSD) + Modem + PAD$	5 930
	$LDC_{D6_warm} = (SSU_{D6} + HDLC-CB + HDLC-M-Modul + PSD) + Modem + PAD$	8 150
	$LDC_{D10_warm} = (SSU_{D10} + HDLC-CB + HDLC-M-Modul + PSD_{D10}) + Modem + PAD$	6 911
LDC _{RaSTA}	$LDC_{D6_warm} = (SSU_{D6} + PSD)$	4 980
	$LDC_{D10_warm} = (SSU_{D10} + PSD_{D10})$	3 741

Table 8: Subsystems - Logical equations and failure rates

Note: The FiT value for the LDC_{RCU} warm standby subsystem is the same as in a non-redundant case, because during switchover the communication will be interrupted for a short time. So, this affects the MTBF value. On the other hand, the switchover time (< 1 min) to the redundant unit is very short compared to an MTTR of 8 hours. Therefore, the LDC_{RCU} warm standby subsystem has a much better *availability* than a non-redundant LDC_{RCU} subsystem. Since a failure in an LDC_{RCU} warm standby subsystem is operationally relevant only in case the primary component fails, only the failure rate of the active hardware is considered in the calculations above. The same applies for the LDC_{RaSTA} warm standby subsystem.

6.9 Results of the analysis

In the following table the MTBF and availability figures of the subsystems are listed.

Subsystem		FiT [1E-9/h] GB 40	MTBF [h]	MTBF [y]	Availability [%]	MTTR
CC	CC _{D2}	4 173	2.40E+05	27	99.999 979 1	1/μ _{SO} = 3 min 1/μ = 8 h
	CC _{D6}	6 393	1.56E+05	18	99.999 968 0	
	CC _{D10}	4 700	2.13E+05	24	99.999 976 5	
EC	EC _{D2}	0.58	1.73E+09	197 988	99.999 999 5	1/μ = 8 h
	EC _{D6}	1.08	9.24E+08	105 515	99.999 999 1	
	EC _{D10}	0.68	1.47E+09	167 321	99.999 999 5	
EC-MUX	EC-MUX _{D6}	0.65	1.53E+09	174 569	99.999 999 5	1/μ = 8 h
	EC-MUX _{D10}	0.35	2.83E+09	322 983	99.999 999 7	
IC	IC1D	2 720	3.68E+05	42	99.997 824 0	1/μ = 8 h
	IMO3D	2 642	3.79E+05	43	99.997 886 4	
	PMO	1 856	5.39E+05	62	99.998 515 2	
MMI	WP ₁₀₀₁	10 000	1.00E+05	11	99.992 000 6	1/μ = 8 h
	WP ₁₀₀₂	1.60	6.25E+08	71 347	99.999 998 7	1/μ = 8 h
	WP _{1002_so}	2 002	5.00E+05	57	99.999 990 0	1/μ = 3 min
Comm. Media	CM ₁₀₀₂	0.08	1.29E+10	1 474 112	99.999 999 9	1/μ = 8 h
Comm. Media remote	CM_rem ₁₀₀₂	0.44	2.26E+09	257 757	99.999 999 6	1/μ = 8 h
LDC _{TCD}	LDC _{D2_hot}	0.77	1.30E+09	148 563	99.999 999 4	1/μ = 8 h
	LDC _{D6_hot}	1.34	7.47E+08	85 218	99.999 998 9	
	LDC _{D10_hot}	1.00	9.99E+08	114 002	99.999 999 2	
LDC _{RCU}	LDC _{D2_warm}	5 930	1,69E+05	19	99.999 990 1	1/μ _{SO} = 1 min 1/μ = 8 h
	LDC _{D6_warm}	8 150	1,23E+05	14	99.999 986 4	
	LDC _{D10_warm}	6 911	1,45E+05	17	99.999 988 5	
LDC _{RaSTA}	LDC _{D6_warm}	4 980	2,01E+05	23	99,999 991 7	1/μ _{SO} = 1 min 1/μ = 8 h
	LDC _{D10_warm}	3 741	2,67E+05	31	99,999 993 8	

Table 9: Subsystems - MTBF and availability

Note: There are also LDC_{RCU} variants with RCU and SCU residing on more than one MEN-D2/MEN-D6/MEN-D10 board. In this case the FiT value is raised by 2 760 FiT (MEN-D2 + PSD), by 4 980 FiT (MEN-D6 + PSD) resp. by 3 741 FiT (MEN-D10 + PSD_{D10}).

The FiT values for the LDC_{RaSTA} variants in warm standby are significantly lower than the LDC_{RCU} variants since the HW components HDLC and PAD Multiprotocol Switch are not required for this type of connection.

7 System analysis

The system to be examined consists of a communication media with two switches, a CC subsystem in warm standby mode and four up to eight hot standby EC/EC-MUX subsystems. A conservative approach is assumed by not differentiating between different severity levels of failures. Each random HW failure contributes to an overall system failure.



According to [EL2_SYS_LIM_AT] the maximum number of EC to be managed by the vital CC core is eight. This also includes mixed EC/EC-MUX architectures installed on central stations resp. remote stations.

Remotely installed EC/EC-MUX are always MEN-D6 or MEN-D10. Central stations may be equipped with either MEN-D2, MEN-D6 or MEN-D10 based technology.

The EC-MUX provides the same behaviour as the EC with respect to dependability. The focus in this analysis is on architectural variants of local EC and remote EC. A closer look of EC-MUX from RAM point of view can be found in [RAM_ANA_SEA].

Note: A failure of the LDC does not affect the core system consisting of CC and EC/EC-MUX. Therefore it is examined separately.

7.1 Results of the analysis

In this section dependability figures for ELEKTRA 2.x variants are listed according to the subsystem analysis above. The variants considered here cover configurations from four up to eight EC/EC-MUX subsystems, ranging from medium to maximum required sizes. Among these are typical variants with one resp. two remotely controlled EC/EC-MUX and a maximum variant with seven remotely controlled EC.

The first part contains MEN-D2 based vital core technology in the central station, the second part MEN-D6 and the third part MEN-D10 based technology.

$$SYSTEM = CC + CM + k EC + m CM_{rem} + m EC_{rem} + n EC-MUX_{rem}$$

Please note that according to the constraints listed in 6.4.4 following relations in the equality above hold:

- $k \geq 1$
- $1 \leq k + m + n \leq 8$
- $n \leq m$

System	FiT [1E-9/h] GB 40	MTBF [h]	MTBF [y]	Availability [%]	MTTR
System_{4i_D2} = CC_{D2} + CM + 4 EC_{D2}	4 175	2.39E+05	27.3	99.999 977 2	1/μ _{SO} = 3 min 1/μ = 8 h
System_{8i_D2} = CC_{D2} + CM + 8 EC_{D2}	4 178	2.39E+05	27.3	99.999 975 4	
System_{4i_D2_1rem_1e} = CC_{D2} + CM + 4 EC_{D2} + CM_{rem} + EC_{rem_D6} + EC_MUX_{rem_D6}	4 178	2.39E+05	27.3	99.999 975 5	
System_{4i_D2_2rem_4e} = CC_{D2} + CM + 4 EC_{D2} + 2 CM_{rem} + 2 EC_{rem_D6} + 2 EC_MUX_{rem_D6}	4 180	2.39E+05	27.3	99.999 973 7	
System_{1i_D2_7rem_7e} = CC_{D2} + CM + 1 EC_{D2} + 7 CM_{rem} + 7 EC_{rem_D6}	4 184	2.39E+05	27.3	99.999 970 1	
System_{D6} = CC_{D6} + CM + 4 EC_{D6}	6 397	1.56E+05	17.8	99.999 964 5	1/μ _{SO} = 3 min 1/μ = 8 h
System_{D6} = CC_{D6} + CM + 8 EC_{D6}	6 402	1.56E+05	17.8	99.999 961 0	
System_{4i_1rem_2e} = CC_{D6} + CM + 4 EC_{D6} + CM_{rem} + EC_{rem_D} + EC_MUX_{rem_D6}	6 400	1.56E+05	17.8	99.999 962 8	
System_{4i_2rem_4e} = CC_{D6} + CM + 4 EC_{D6} + 2 CM_{rem} + 2 EC_{rem_D6} + 2 EC_MUX_{rem_D6}	6 402	1.56E+05	17.8	99.999 961 0	
System_{1i_7rem_7e} = CC_{D6} + CM + 1 EC_{D6} + 7 CM_{rem} + 7 EC_{rem_D6}	6 405	1.56E+05	17.8	99.999 958 6	
System_{D10} = CC_{D10} + CM + 4 EC_{D10}	4 703	2.13E+05	24.3	99.999 974 3	

System	FiT [1E-9/h] GB 40	MTBF [h]	MTBF [y]	Availability [%]	MTTR
System_{D10} = CC_{D10} + CM + 8 EC_{D10}	4 706	2.13E+05	24.3	99.999 972 1	1/μ _{SO} = 3 min 1/μ = 8 h
System_{4i_1rem_2e} = CC_{D10} + CM + 4 EC_{D10} + CM_{rem} + EC_{rem_D10} + EC_MUX_{rem_D10}	4 704	2.13E+05	24.3	99.999 973 1	
System_{4i_2rem_4e} = CC_{D10} + CM + 4 EC_{D10} + 2 CM_{rem} + 2 EC_{rem_D10} + 2 EC_MUX_{rem_D10}	4 706	2.13E+05	24.3	99.999 971 9	
System_{1i_7rem_7e} = CC_{D10} + CM + 1 EC_{D10} + 7 CM_{rem} + 7 EC_{rem_D10}	4 709	2.12E+05	24.2	99.999 969 6	

Table 10: System - MTBF and availability

Subsystem	FiT [1E-9/h] GB 40	MTBF [h]	MTBF [y]	Availability [%]	MTTR
LDC_{D2_hot}	0.77	1.30E+09	148 563	99.999 999 4	1/μ = 8 h
LDC_{TCD} LDC_{D6_hot}	1.34	7.47E+08	85 218	99.999 998 9	
LDC_{D10_hot}	1.00	9.99E+08	114 002	99.999 999 2	
LDC_{RCU} LDC_{D2_warm}	5 930	1.69E+05	19	99.999 990 1	1/μ _{SO} = 1 min 1/μ = 8 h
LDC_{D6_warm}	8 150	1.23E+05	14	99.999 986 4	
LDC_{D10_warm}	6 911	1.45E+05	17	99.999 988 5	
LDC_{RaSTA} LDC_{D6_warm}	4 980	2,01E+05	23	99,999 991 7	1/μ _{SO} = 1 min 1/μ = 8 h
LDC_{D10_warm}	3 741	2,67E+05	31	99,999 993 8	

Table 11: LDC - MTBF and availability

7.2 Fulfillment of requirements

The following requirements are stated in [EL2_SYS_SRS].

- **DPR:0201 (TRT285):** System reliability (System down time < 3 min)
Synopsis: System failures leading to a system down time < 3 minutes: MTBF of the entire system must exceed 10 years.
- **DPR:0202 (TRT286):** System reliability (System down time < 8 h)
Synopsis: System failures leading to a system down time > 3 min < 8 hours: MTBF of the entire system must exceed 2000 years.

These requirements lead to following classification of the required availability of the ELEKTRA 2.x system:

	Duration of repair	Type of repair	MTBF [years]	Availability [%]	Mean downtime per year [secs]
System classified	> 3 min < 8 h	manually	2 000	99.999 95	14
	< 3 min	automatically	10	99.999 94	18

Table 12: Availability classes for ELEKTRA 2.x



The requirements are coupled and the ideas behind their formulation are:

- **Balanced partitioning of availability:** Essentially, for both categories an availability of 99.999 95 % is required. This implies a downtime per year of about 15 seconds for each category.
- **System availability of 99.999 9 %.** The tolerable mean downtime per year is therefore about 30 seconds. So, we have a required availability of 99.999 9 % for the system ELEKTRA 2.x.

So, with focus on the availability aspects of the system, the requirements are

	Duration of repair	Type of repair	Availability [%]	Mean downtime per year [secs]
System	< 8 h	either manually or automatically	99.999 9	30
System classified	> 3 min < 8 h	manually	99.999 95	15
	< 3 min	automatically	99.999 95	15

Table 13: Availability requirements for ELEKTRA 2.x

Comparison with **Table 10: System - MTBF and availability** shows that the system availability of ELEKTRA 2.x fulfills the requirements for all variants.

Result: Requirement in all typical situations fulfilled. This includes fulfillment for up to eight EC locally installed and the maximum allowed EC/EC-MUX configuration with two remote stations.

- **DPR:5207 (TRT833):** Serial connection ELEKTRA <-> AzLM - No Availability Restrictions
Synopsis: The serial connection ELEKTRA <-> AzLM must keep the availability of the eStW within the RAM requirements.
 - The connection is realised via redundant serial links between one redundant EC and the AzLM. Each redundancy of this EC is connected with one serial link to the AzLM. A link failure of one instance is tolerated. It has no operational consequences.
 - The EC communicates via OCS-SB, an application residing on the EC HW board, with AzLM. An outage of the OCS-SB application instance has the same consequences as an outage of the EC application, leading to a restart of the HW board. For more information about these aspects see [IF_EL2_AzLM_ADD].

From HW reliability point of view an EC subsystem with a connection to AzLM shows the same failure behavior as an EC subsystem without connection to AzLM.

Result: Requirement fulfilled

END OF DOCUMENT